E-Commerce Customer Churn Prediction & Analytics

Author: Machine Learning Engineering Team
Project Status: Completed & Ready for Deployment
Target Metric: Recall (Churn Class) & ROC-AUC

---

1. Executive Summary

Customer churn represents one of the most critical challenges for e-commerce platforms, directly impacting recurring revenue and Customer Lifetime Value (CLV). This project delivers an end-to-end Machine Learning solution designed to proactively identify customers at high risk of churning before they discontinue their engagement with the platform.
Through rigorous experimentation, feature selection, and hyperparameter tuning, the final **Tuned Random Forest Model** achieved exceptional predictive performance on unseen test data:

- **Recall (Churn = 1): 88.0%** — Successfully identifying 164 out of 186 churning customers in the test evaluation.
- **ROC-AUC Score: 0.9925** on test data (0.9943 Cross-Validation score).
- **Overall Accuracy: 96.58%**.
- **Key Insight:** Customer churn is predominantly driven by declining engagement (39.5%) and recency of last purchase (28.2%), rather than customer support ticket volume or discount usage.

2. Machine Learning Pipeline & Methodology

The project followed a structured four-phase pipeline to ensure reproducibility, prevent data leakage, and maximize model interpretability:

1. **Data Preprocessing & Stratified Splitting:** The dataset was split using an 80/20 train-test ratio with stratification to preserve class distributions (approximately 15.5% churn prevalence). Standard scaling was applied exclusively after splitting.
2. **Baseline Modeling (Phase 1):** Evaluated multiple baseline models—Logistic Regression, Random Forest, and Decision Tree—across all 10 initial features.
3. **Feature Selection (Phase 2):** Conducted feature importance analysis to prune redundant or noisy features, comparing 10-feature vs. 8-feature models.
4. **Hyperparameter Tuning (Phase 3):** Applied GridSearchCV with 5-fold cross-validation and class weight balancing to optimize recall for the minority class.
5. **Explainability & Business Mapping (Phase 4):** Extracted feature importance scores to translate technical outputs into strategic retention recommendations.

3. Model Performance & Experimental Progression

Phase 1: Baseline Comparison (Top 10 Features)

Initial evaluation across all 10 features provided a benchmark for model selection:

Model	Accuracy	ROC-AUC	Precision (Churn=1)	Recall (Churn=1)	False Negatives
Logistic Regression	96.67%	0.9940	0.91	0.87	25
Random Forest	96.25%	0.9930	0.91	0.84	29
Decision Tree	95.67%	0.9173	0.86	0.86	26

Phase 2: Feature Selection Impact (Top 10 vs. Top 8)

Pruning the two lowest-impact features (customer support tickets and discount usage rate) reduced model complexity and eliminated noise without sacrificing accuracy:

Setup	Accuracy	ROC-AUC	Precision (Churn=1)	Recall (Churn=1)	F1-Score
Random Forest (Top 10)	96.25%	0.9930	0.91	0.84	0.87
Random Forest (Top 8)	**96.83%**	**0.9926**	**0.94**	**0.85**	**0.89**

Phase 3: Hyperparameter Tuning (GridSearchCV Results)

By leveraging class weighting ('class_weight': 'balanced') and optimizing tree depth and split criteria, the model significantly boosted its recall score:

Metric	Baseline RF (Top 8)	Tuned RF (Phase 3)	Impact / Value Added
Accuracy	96.83%	96.58%	Slight, negligible trade-off
ROC-AUC Score	0.9926	0.9925	Consistent high generalization
Recall (Churn)	0.85	**0.88**	**+3.0% boost in detecting at-risk churners**
Precision (Churn)	0.94	0.90	Balanced trade-off

Metric	Baseline RF (Top 8)	Tuned RF (Phase 3)	Impact / Value Added
			for higher recall

Optimal Hyperparameters Identified:

```
{
    'class_weight': 'balanced',
    'max_depth': 20,
    'min_samples_leaf': 1,
    'min_samples_split': 5,
    'n_estimators': 200
}
```

4. Feature Importance & Business Interpretability

Feature importance analysis revealed that customer engagement and purchasing recency dictate nearly 90% of the prediction outcome:

Rank	Feature Name	Importance Score	Business Domain Meaning
1	engagement_score	**39.45%**	Platform activity level & interaction frequency
2	days_since_last_purchase	**28.18%**	Recency of transaction activity
3	inactivity_risk_score	**21.52%**	Composite risk indicator calculated from user sessions
4	satisfaction_score	4.31%	User rating & feedback sentiment
5	product_review_score_avg	2.34%	Product feedback score
6	customer_support_tickets	1.69%	Support ticket count
7	discount_usage_rate	1.57%	Discount redemption behavior
8	loyalty_member	0.93%	Loyalty tier flag

5. Strategic Business Recommendations

- **Automated Retention Alerts:** Integrate the API into the CRM system to flag customers whose engagement_score drops below critical thresholds or whose inactivity_risk_score spikes.
- **Proactive Re-engagement Campaigns:** Target customers approaching high days_since_last_purchase values with personalized product recommendations before they decide to leave.
- **Optimized Discount Allocation:** Reallocate marketing budget away from blanket discounts towards high-value customers identified by the churn model.

6. Technical Artifacts & Deployment Readiness

All pipeline artifacts have been serialized using joblib and are ready for deployment via a REST API (e.g., FastAPI) or interactive web interface (e.g., Streamlit):

- best_churn_model_rf.pkl — Saved Tuned Random Forest Model
- scaler_top8.pkl — Fitted StandardScaler for input transformation
- top_8_features_list.pkl — Schema list for input validation